

# Ellipses and Hyperbolas: Reading a Model's Key Features

## Answer Key

### Algebra 2

#### Quick Answers

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|----------------------------|----------------------------|
| 1. 5                       | 6. 6, 64                   |
| 2. 6, 10                   | 7. 2, 5.39                 |
| 3. 8, -2, 7, -1            | 8. $m = 45$ , $n = 20$ , 5 |
| 4. $\frac{3}{4}$           | 9. $\frac{7}{4}$           |
| 5. $a = 7000$ , $c = 2800$ | 10. $\frac{5}{6}$          |
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#### Curriculum

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**Level:** Algebra 2

HSG-GPE.A.3 – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Difficulty 1–5 of 5 across 18 tagged checks.

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#### Common wrong answers

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9. If they got 0.57: divided the  $x$ -denominator by the  $y$ -denominator, which reverses the slope

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**1. Whispering gallery:**  $\frac{x^2}{169} + \frac{y^2}{144} = 1$  (feet). How far from the centre is each focus?

**Solution:** The larger denominator sits under  $x^2$ , so the major axis is horizontal with  $a^2 = 169$  and  $b^2 = 144$ . For an ellipse the focal relation is  $c^2 = a^2 - b^2$  (subtract, because the foci lie inside the curve):  $c^2 = 169 - 144 = 25$ , so  $c = \sqrt{25}$ . Each focus is that far from the centre, one on each side.

$c = 5$  ft

**2. Radio towers:**  $\frac{x^2}{36} - \frac{y^2}{64} = 1$  (kilometres). Give the vertices and the foci.

**Solution:** The  $x^2$  term is positive, so the transverse axis is horizontal:  $a^2 = 36$  and  $b^2 = 64$ . The vertices are  $(\pm a, 0)$  with  $a = \sqrt{36}$ .

$(\pm 6, 0)$

For a hyperbola the focal relation adds:  $c^2 = a^2 + b^2 = 36 + 64 = 100$ , so  $c = \sqrt{100}$  and the foci are  $(\pm c, 0)$ .

$(\pm 10, 0)$

Sanity check:  $c > a$ , as it must be — the foci always sit beyond the vertices on a hyperbola.

**3. Sprinkler patch:**  $\frac{(x-3)^2}{25} + \frac{(y+2)^2}{9} = 1$  (metres). Give the endpoints of the major axis, then the foci.

**Solution:** The centre is  $(h, k) = (3, -2)$ , read by flipping the signs inside the parentheses. The larger denominator, 25, sits under the  $x$  term, so the major axis is horizontal with  $a = \sqrt{25} = 5$  and  $b = \sqrt{9} = 3$ . Step 5 metres right and left of the centre:  $3 + 5$  and  $3 - 5$ , each paired with  $y = -2$ .

$$(8, -2)$$

$$(-2, -2)$$

Focal distance:  $c^2 = a^2 - b^2 = 25 - 9 = 16$ , so  $c = 4$ . Step 4 metres right and left of the centre along the same axis:  $3 + 4$  and  $3 - 4$ .

$$(7, -2)$$

$$(-1, -2)$$

**4. Cooling tower:**  $\frac{x^2}{144} - \frac{y^2}{81} = 1$  (metres). Find the asymptote slopes and equations.

**Solution:** Here  $a = \sqrt{144} = 12$  and  $b = \sqrt{81} = 9$ . For a hyperbola opening left and right, the asymptotes through the centre have slopes  $\pm \frac{b}{a} = \pm \frac{9}{12}$ , which reduces.

$$\frac{3}{4}$$

The centre is the origin, so the two asymptote equations are  $y = \frac{3}{4}x$  and  $y = -\frac{3}{4}x$ ; together

$$y = \pm \frac{3}{4}x$$

**5. Satellite orbit:** closest approach 4200 km, farthest 9800 km, with the planet at one focus. Solve for  $a$  and  $c$ .

**Solution:** The closest point is the vertex nearer the focus, a distance  $a - c$  away; the farthest is the opposite vertex,  $a + c$  away. That is the system

$$a - c = 4200$$

$$a + c = 9800$$

Add the two equations to eliminate  $c$ :  $2a = 14000$ , so  $a = 7000$  km.

$$a = 7000$$

Substitute back into the second equation:  $7000 + c = 9800$ , so  $c = 2800$  km.

$$c = 2800$$

Check:  $7000 - 2800 = 4200$  ✓ and  $7000 + 2800 = 9800$  ✓.

**6. Navigation stations:** foci 20 km apart, distance difference 12 km. Find  $a$ , then  $b^2$ .

**Solution:** On a hyperbola the constant difference of distances to the foci equals  $2a$ , so  $2a = 12$  and  $a = \frac{12}{2}$ .

$$a = 6$$

The stations are the foci, so  $2c = 20$  and  $c = 10$ . The hyperbola relation  $c^2 = a^2 + b^2$  rearranges to  $b^2 = c^2 - a^2 = \left(\frac{20}{2}\right)^2 - \left(\frac{12}{2}\right)^2 = 100 - 36$ .

$$b^2 = 64$$

**7. Hyperbolic mirror:** vertices  $(\pm 5, 0)$  cm, asymptotes  $y = \pm \frac{2}{5}x$ . Find  $b$  and  $c$ .

**Solution:** The vertices give  $a = 5$ . The asymptote slope of a left-right hyperbola is  $\frac{b}{a}$ , so

$$\frac{b}{5} = \frac{2}{5}, \text{ which means } b = 5 \cdot \frac{2}{5}.$$

$$b = 2$$

Then  $c^2 = a^2 + b^2 = 5^2 + 2^2 = 29$ , so  $c = \sqrt{29} = 5.385 \dots$  centimetres, which rounds to two decimal places.

$$c \approx 5.39$$

**8. Bridge arch:** upper half of an ellipse centred at the origin through  $(3, 4)$  and  $(6, 2)$  (metres). Find  $m = a^2$ ,  $n = b^2$ , and the foci.

**Solution:** Substituting each point into  $\frac{x^2}{m} + \frac{y^2}{n} = 1$  gives

$$\begin{aligned}\frac{9}{m} + \frac{16}{n} &= 1 \\ \frac{36}{m} + \frac{4}{n} &= 1\end{aligned}$$

These are linear in  $\frac{1}{m}$  and  $\frac{1}{n}$ . Multiply the second equation by 4 and subtract the first:  $\frac{144}{m} - \frac{9}{m} = 4 - 1$ , so  $\frac{135}{m} = 3$  and  $m = 45$ .  $m = 45$

Substitute into the first equation:  $\frac{9}{45} + \frac{16}{n} = 1$  gives  $\frac{16}{n} = \frac{4}{5}$ , so  $n = 20$ .  $n = 20$

Since  $m > n$ , the major axis is horizontal and  $c^2 = m - n = 45 - 20 = 25$ , so  $c = 5$  metres.  $(\pm 5, 0)$

**9. Tunnel, error analysis:**  $\frac{y^2}{49} - \frac{x^2}{16} = 1$  (metres). Jae reports slopes  $\pm 0.57$ . Find the mistake and the correct slopes.

**Solution:** This hyperbola has the  $y^2$  term positive, so it opens up and down and its transverse axis is vertical:  $a^2 = 49$  sits under  $y^2$  and  $b^2 = 16$  sits under  $x^2$ . For an up-down hyperbola the asymptotes are  $y = \pm \frac{a}{b}x$  — the number from under  $y^2$  goes on top. Jae used

$\frac{4}{7} = 0.57$ , which is  $\frac{b}{a}$ : the slope was inverted by copying the left-right formula. The correct slope is  $\frac{\sqrt{49}}{\sqrt{16}} = \frac{7}{4}$ .  $y = \pm \frac{7}{4}x$

Reading check: the tunnel is taller than it is wide near the vertex, so its walls must rise faster than 1 metre up per 1 metre across — a slope bigger than 1, not smaller.

**10. Skate ramp:**  $\frac{x^2}{36} - \frac{y^2}{25} = 1$  (metres). Asymptote slope, sketch, and explanation.

(a)  $a = \sqrt{36} = 6$  and  $b = \sqrt{25} = 5$ , so the slopes are  $\pm \frac{b}{a}$ .  $\frac{5}{6}$

(b) Model sketch: mark the vertices at  $(\pm 6, 0)$ ; draw the central rectangle with corners  $(\pm 6, \pm 5)$ ; draw the two diagonals of that rectangle all the way out — those are  $y = \pm \frac{5}{6}x$ ; then draw one branch opening right from  $(6, 0)$  and one opening left from  $(-6, 0)$ , each hugging the diagonals as it goes.

(c) Model explanation (judge by reading): solving the equation for  $y$  gives  $y = \pm \frac{5}{6}\sqrt{x^2 - 36} = \pm \frac{5}{6}x\sqrt{1 - \frac{36}{x^2}}$ . As  $|x|$  grows,  $\frac{36}{x^2}$  shrinks toward 0, so the square-root factor creeps up toward 1 and the branch creeps up toward  $\pm \frac{5}{6}x$ . It never arrives: the factor is always strictly less than 1 because  $\frac{36}{x^2}$  is never 0 for a finite  $x$ , so the branch stays strictly inside its asymptote. Equivalently, a point on the asymptote would force  $x^2/36 - y^2/25 = 0$ , not 1.